

Assignment 02

Analysis of Algorithms

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GitHub Repository:

<https://github.com/darioXxx32/Algorithm-analysis/tree/main/assignment02>

Exercise 1: Solving Recurrences

1.1. $T(n) = 3T(n/2) + n$, $T(1) = 0$, for $n = 2^k$

Primary handwritten resolution:

Assignment 02k

1. Solve the recurrence:

for all powers of two 1, 2, 4, 8, 16

1.1) $T(n) = 3T(n/2) + n$ $T(1) = 0$

$T(n) = 3T(n/2) + n$
 * Substitute $T(n/2) = 3T(n/4) + n/2$
 $T(n) = 3[3T(n/4) + n/2] + n$
 $T(n) = 3^2 T(n/4) + \frac{3}{2}n + n$
 * Substitute $T(n/4) = 3T(n/8) + n/4$
 $T(n) = 3^2 [3T(n/8) + \frac{n}{4}] + \frac{3}{2}n + n$
 $T(n) = 3^3 (n/8) + (\frac{3}{2})^2 n + \frac{3}{2}n + n$

Following $\log$ steps: We observed pattern after K substitutions, we can express the recurrence as:

$T(n) = 3^K T(n/2^K) + n \sum_{i=0}^{K-1} (\frac{3}{2})^i$

We are given that n is a power of two ($n=2^K$) which implies $K = \log_2 n$, when we reach the reach case $T(1)$ the terms inside the function become 1, meaning 1, meaning $\frac{n}{2^K} = 1$

substitute $T(1) = 0$ into our generalized eq.

$T(n) = 3^K (0) + n \sum_{i=0}^{K-1} (\frac{3}{2})^i$
 $T(n) = n \sum_{i=0}^{K-1} (\frac{3}{2})^i$

now we need remembered

$\sum_{i=0}^{K-1} r^i = \frac{r^K - 1}{r - 1}$ when $r = \frac{3}{2}$

$T(n) = n \left[\frac{(\frac{3}{2})^K - 1}{\frac{3}{2} - 1} \right]$
 $T(n) = n \left[\frac{(\frac{3}{2})^K - 1}{1/2} \right]$
 $T(n) = 2n \left(\frac{3^K}{2^K} - 1 \right)$ and $n=2^K$
 $T(n) = 2n \left(\frac{3^K}{n} - 1 \right)$
 $T(n) = 2n \left(3^K - n \right)$

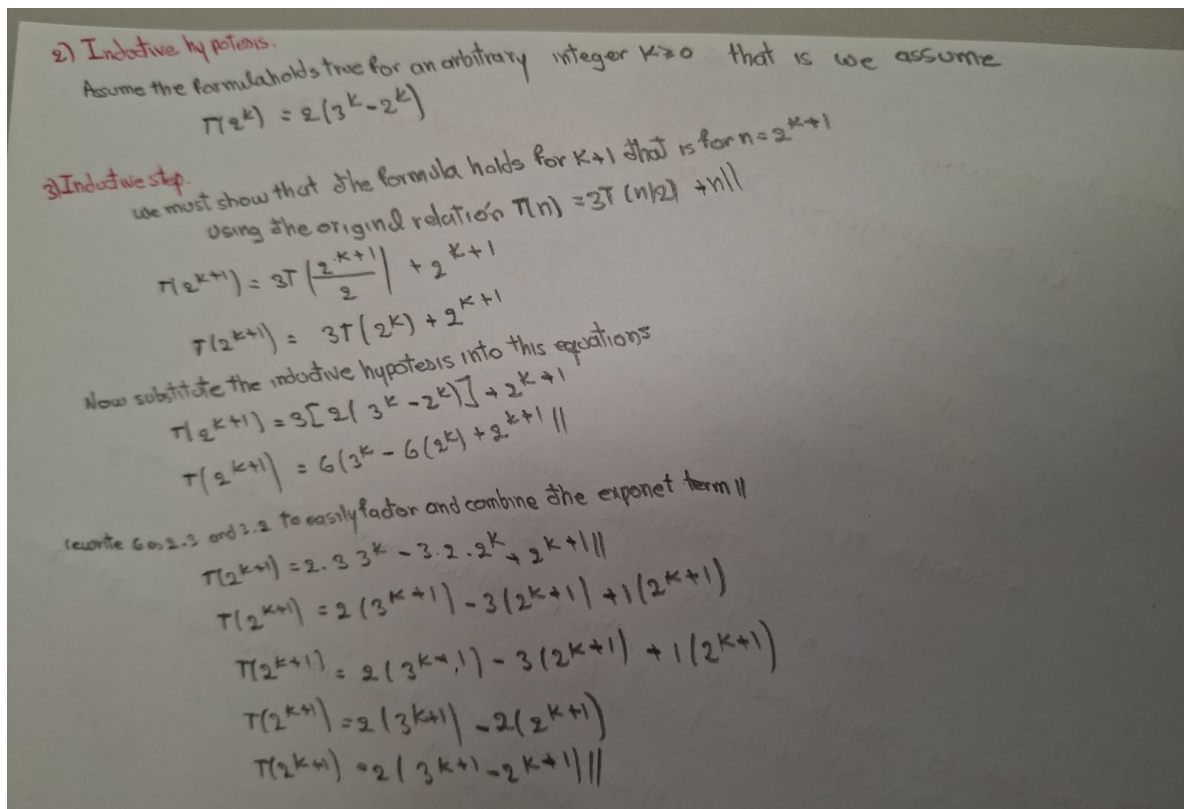
and now recall $K = \log_2 n$

$3^K = 3^{\log_2 n} = n^{\log_2 3}$
 $T(n) = 2(n^{\log_2 3} - n)$

Part 2) Verifying using Mathematical Induction

i) Base Case ($K=0$)
 For $K=0$, we have $n=2^0=1$
 using our closed-form formula
 $T(2^0) = 2(3^0 - 2^0) = 2(1-1) = 0$, the base case holds

Exercise 1.1 handwritten work: substitutions, pattern, and geometric sum.



Exercise 1.1 handwritten work: induction verification.

Typed solution for contrast and verification.

Part 1: Finding the Closed Form by Repeated Substitution.

Starting with

$$T(n) = 3T(n/2) + n,$$

we substitute the recursive term several times to see the pattern:

$$\begin{aligned}
 T(n) &= 3T(n/2) + n \\
 &= 3 \left[3T(n/4) + \frac{n}{2} \right] + n \\
 &= 3^2 T(n/4) + \frac{3}{2}n + n \\
 &= 3^2 \left[3T(n/8) + \frac{n}{4} \right] + \frac{3}{2}n + n \\
 &= 3^3 T(n/8) + \left(\frac{3}{2} \right)^2 n + \frac{3}{2}n + n.
 \end{aligned}$$

After k substitutions, the recurrence has the form

$$T(n) = 3^k T(n/2^k) + n \sum_{i=0}^{k-1} \left(\frac{3}{2} \right)^i$$

Since n is a power of two, $n = 2^k$, so $k = \log_2 n$. At this point $n/2^k = 1$, and the base case

gives $T(1) = 0$:

$$\begin{aligned}
T(n) &= 3^k(0) + n \sum_{i=0}^{k-1} \left(\frac{3}{2}\right)^i \\
&= n \left[\frac{(3/2)^k - 1}{(3/2) - 1} \right] \\
&= n \left[\frac{(3/2)^k - 1}{1/2} \right] \\
&= 2n \left(\frac{3^k}{2^k} - 1 \right).
\end{aligned}$$

Using $2^k = n$, this becomes

$$T(n) = 2n \left(\frac{3^k}{n} - 1 \right) = 2(3^k - n).$$

Finally, since $k = \log_2 n$,

$$T(n) = 2(3^{\log_2 n} - n) \quad \text{or equivalently} \quad T(2^k) = 2(3^k - 2^k).$$

Part 2: Verification by Mathematical Induction.

We prove that

$$T(2^k) = 2(3^k - 2^k)$$

for all integers $k \geq 0$.

- **Base Case** ($k = 0$): Here $n = 2^0 = 1$. The formula gives

$$T(2^0) = 2(3^0 - 2^0) = 2(1 - 1) = 0,$$

which matches the condition $T(1) = 0$.

- **Inductive Hypothesis:** Assume that for an arbitrary integer $k \geq 0$,

$$T(2^k) = 2(3^k - 2^k).$$

- **Inductive Step:** We prove the formula for $k + 1$. Since $\frac{2^{k+1}}{2} = 2^k$, the recurrence gives

$$\begin{aligned}
T(2^{k+1}) &= 3T(2^k) + 2^{k+1} \\
&= 3[2(3^k - 2^k)] + 2^{k+1} \\
&= 2 \cdot 3 \cdot 3^k - 3 \cdot 2 \cdot 2^k + 2^{k+1} \\
&= 2 \cdot 3^{k+1} - 3 \cdot 2^{k+1} + 2^{k+1} \\
&= 2 \cdot 3^{k+1} - 2 \cdot 2^{k+1} \\
&= 2(3^{k+1} - 2^{k+1}).
\end{aligned}$$

Therefore the closed form is verified.

1.2. $T(n) = 3T(n-1) + 4T(n-2)$, $T(0) = 0$, $T(1) = 5$

Primary handwritten resolution:

1.2) $T(n) = 3T(n-1) + 4T(n-2)$ with initial condition $T(0) = 0$, $T(1) = 5$
 identify the given equation is $T(n) = 3T(n-1) + 4T(n-2)$ with initial condition $T(0) = 0$, $T(1) = 5$
 Second linear order homogeneous.
 $T(n) = r^n$

$$\frac{r^n}{n-2} = \frac{3r^{n-1}}{n-2} + \frac{4r^{n-2}}{n-2}$$

$$= r^2 = 3r + 4 \Rightarrow (r^2 - 3r - 4) = 0 \Rightarrow \begin{cases} r_1 = 4 \\ r_2 = -1 \end{cases}$$

$$C_2 = -C_1$$

$$5 = 4C_1 - (-C_1)$$

$$5 = 5C_1$$

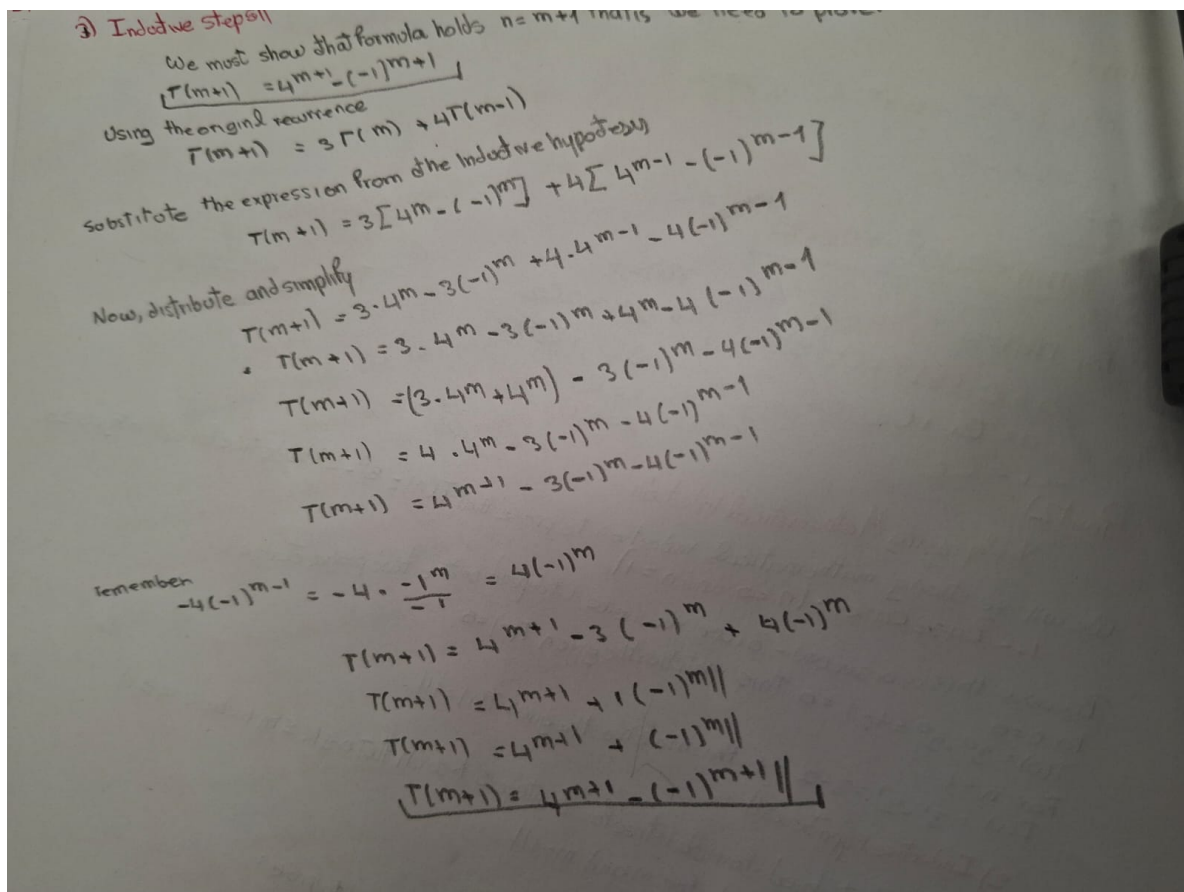
$$1 = C_1$$

$$-1 = C_2 \text{ therefore } (4)^n - 1(-1)^n$$

$$4^n - (-1)^n$$
 Part 2) Verify by math induction
 $T(n) = 4^n - (-1)^n$ holds that it is true for all integer $n \geq 0$
 Base Case
 $n = 0$
 $T(0) = 4^0 - (-1)^0 = 1 - 1 = 0$ $T(0) = 0$ ✓
 $T(1) = 4^1 - (-1)^1 = 4 - (-1) = 5$ $T(1) = 5$ ✓
 2) Inductive hypothesis.
 Assume the proposed formula is true as true for all integers k such that $0 \leq k \leq m$ (where $m \geq 1$)
 Specifically for m and $m-1$

$$\begin{cases} T(m) = 4^m - (-1)^m \\ T(m-1) = 4^{m-1} - (-1)^{m-1} \end{cases}$$

Exercise 1.2 handwritten work: characteristic equation, constants, and base cases.



Exercise 1.2 handwritten work: inductive step.

Typed solution for contrast and verification.

Part 1: Finding the Closed Form.

This is a second-order linear homogeneous recurrence. Following the handwritten solution, assume a solution of the form $T(n) = r^n$. Then

$$r^n = 3r^{n-1} + 4r^{n-2}.$$

Dividing by r^{n-2} gives the characteristic equation

$$r^2 = 3r + 4 \implies r^2 - 3r - 4 = 0.$$

Factoring,

$$(r - 4)(r + 1) = 0,$$

so the roots are $r_1 = 4$ and $r_2 = -1$. Therefore,

$$T(n) = c_1 4^n + c_2 (-1)^n.$$

Now apply the initial conditions:

$$\begin{aligned}T(0) = c_1 + c_2 = 0 &\implies c_2 = -c_1, \\T(1) = 4c_1 - c_2 = 5.\end{aligned}$$

Substituting $c_2 = -c_1$ into the second equation:

$$\begin{aligned}5 &= 4c_1 - (-c_1) \\&= 5c_1,\end{aligned}$$

so $c_1 = 1$ and $c_2 = -1$. Thus the closed form is

$$T(n) = 4^n - (-1)^n$$

Part 2: Verification by Strong Mathematical Induction.

We prove that

$$T(n) = 4^n - (-1)^n$$

for all integers $n \geq 0$. Strong induction is used because the recurrence depends on the two previous values.

- **Base Cases:**

$$\begin{aligned}T(0) &= 4^0 - (-1)^0 = 1 - 1 = 0, \\T(1) &= 4^1 - (-1)^1 = 4 + 1 = 5.\end{aligned}$$

These match the given initial conditions.

- **Inductive Hypothesis:** Assume the formula is true for all integers k with $0 \leq k \leq m$, where $m \geq 1$. In particular,

$$T(m) = 4^m - (-1)^m \quad \text{and} \quad T(m-1) = 4^{m-1} - (-1)^{m-1}.$$

- **Inductive Step:** We prove the formula for $m+1$:

$$\begin{aligned}T(m+1) &= 3T(m) + 4T(m-1) \\&= 3[4^m - (-1)^m] + 4[4^{m-1} - (-1)^{m-1}] \\&= 3 \cdot 4^m - 3(-1)^m + 4^m - 4(-1)^{m-1}.\end{aligned}$$

Since $(-1)^{m-1} = -(-1)^m$, the last term becomes $+4(-1)^m$. Hence

$$\begin{aligned}T(m+1) &= 3 \cdot 4^m + 4^m - 3(-1)^m + 4(-1)^m \\&= 4 \cdot 4^m + (-1)^m \\&= 4^{m+1} - (-1)^{m+1}.\end{aligned}$$

Therefore the formula is verified.

1.3. $T(n) = 5T(n-1) - 6T(n-2)$, $T(0) = 0$, $T(1) = 1$

Primary handwritten resolution:

$T(n) = 5T(n-1) - 6T(n-2)$ $T(0) = 0$ $T(1) = 1$
 $T(n) = r^n$
 $r^n = 5r^{n-1} - 6r^{n-2}$ $\times r^{n-2}$
 $r^2 = 5r - 6$
 $\begin{cases} r_1 = 3 \\ r_2 = 2 \end{cases}$
 $T(n) = C_1(3)^n + C_2(2)^n$
 For $T(0) = 0$
 $0 = C_1(3)^0 + C_2(2)^0$
 $0 = C_1 + C_2$
 $-C_1 = C_2$
 For $T(1) = 1$
 $1 = 3C_1 + 2C_2$
 $1 = 3C_1 + 2(-C_1)$
 $1 = C_1$
 $-1 = C_2$
 Substitute
 $T(n) = 1(3)^n - 1(2)^n$
 $T(n) = 3^n - 2^n$
 Part 2)
 Verify using Mathematical Induction
 We will use strong mathematical induction to prove that $T(n) = 3^n - 2^n$ holds true for all integers ≥ 0 .
 1. Base Cases: ($n=0$ and $n=1$)
 Because this is a second-order recurrence (it depends on the two previous terms)
 For $n=0$
 $T(0) = 3^0 - 2^0 = 1 - 1 = 0$ this matches the given $T(0) = 0$
 For $n=1$
 $T(1) = 3^1 - 2^1 = 3 - 2 = 1$ this matches the given $T(1) = 1$
 2) Inductive hypothesis)
 Assume that closed-formula is true for all integers k such that $0 \leq k \leq m$ (where $m \geq 1$)
 Specifically, we assume it holds for m and $m-1$)
 $T(m) = 3^m - 2^m$
 $T(m-1) = 3^{m-1} - 2^{m-1}$
 3) Inductive step.
 We must show that the formula holds for $n=m+1$ that is, we need to prove:
 $T(m+1) = 3^{m+1} - 2^{m+1}$
 Using the recurrence relation
 $T(m+1) = 5T(m) - 6T(m-1)$
 Substitute the expression from the inductive hypothesis
 $T(m+1) = 5[3^m - 2^m] - 6[3^{m-1} - 2^{m-1}]$
 Now, distribute the terms.
 $T(m+1) = 5 \cdot 3^m - 5 \cdot 2^m - 6 \cdot 3^{m-1} + 6 \cdot 2^{m-1}$
 to simplify, rewrite the 6 multipliers as $2 \cdot 3$ to match the bases of the exponent terms:
 $T(m+1) = 5 \cdot 3^m - 5 \cdot 2^m - (2 \cdot 3) \cdot 3^{m-1} + (3 \cdot 2) \cdot 2^{m-1}$
 Combine the bases.
 $T(m+1) = 5 \cdot 3^m - 5 \cdot 2^m - 2 \cdot 3^m + 3 \cdot 2^m$
 Group the term with base 3 and the terms with base 2:
 $T(m+1) = (5 \cdot 3^m - 2 \cdot 3^m) - (5 \cdot 2^m - 3 \cdot 2^m)$
 $T(m+1) = (5-2) \cdot 3^m - (5-3) \cdot 2^m$
 $T(m+1) = 3 \cdot 3^m - 2 \cdot 2^m$
 $T(m+1) = 3^{m+1} - 2^{m+1}$

Exercise 1.3 handwritten work: characteristic equation, constants, and induction.

Typed solution for contrast and verification.

Part 1: Finding the Closed Form.

Assume $T(n) = r^n$. Substituting into the recurrence gives

$$r^n = 5r^{n-1} - 6r^{n-2}.$$

Dividing by r^{n-2} ,

$$r^2 = 5r - 6 \implies r^2 - 5r + 6 = 0.$$

Factoring the characteristic equation,

$$(r - 3)(r - 2) = 0,$$

so $r_1 = 3$ and $r_2 = 2$. Therefore,

$$T(n) = c_1 3^n + c_2 2^n.$$

Now use the initial conditions:

$$\begin{aligned} T(0) = c_1 + c_2 &= 0 \implies c_2 = -c_1, \\ T(1) = 3c_1 + 2c_2 &= 1. \end{aligned}$$

Substitute $c_2 = -c_1$:

$$\begin{aligned} 1 &= 3c_1 + 2(-c_1) \\ &= c_1. \end{aligned}$$

Thus $c_1 = 1$ and $c_2 = -1$, so the closed form is

$$T(n) = 3^n - 2^n$$

Part 2: Verification by Strong Mathematical Induction.

We prove that

$$T(n) = 3^n - 2^n$$

for all integers $n \geq 0$.

- **Base Cases:**

$$\begin{aligned} T(0) &= 3^0 - 2^0 = 1 - 1 = 0, \\ T(1) &= 3^1 - 2^1 = 3 - 2 = 1. \end{aligned}$$

These match the given initial conditions.

- **Inductive Hypothesis:** Assume the closed form is true for all integers k such that

$0 \leq k \leq m$, where $m \geq 1$. In particular,

$$T(m) = 3^m - 2^m \quad \text{and} \quad T(m-1) = 3^{m-1} - 2^{m-1}.$$

• **Inductive Step:** We prove the formula for $m+1$:

$$\begin{aligned} T(m+1) &= 5T(m) - 6T(m-1) \\ &= 5[3^m - 2^m] - 6[3^{m-1} - 2^{m-1}] \\ &= 5 \cdot 3^m - 5 \cdot 2^m - 6 \cdot 3^{m-1} + 6 \cdot 2^{m-1}. \end{aligned}$$

Rewrite the last two terms using the same bases:

$$\begin{aligned} 6 \cdot 3^{m-1} &= 2 \cdot 3^m, \\ 6 \cdot 2^{m-1} &= 3 \cdot 2^m. \end{aligned}$$

Therefore,

$$\begin{aligned} T(m+1) &= 5 \cdot 3^m - 5 \cdot 2^m - 2 \cdot 3^m + 3 \cdot 2^m \\ &= (5-2)3^m - (5-3)2^m \\ &= 3 \cdot 3^m - 2 \cdot 2^m \\ &= 3^{m+1} - 2^{m+1}. \end{aligned}$$

Therefore the formula is verified.

1.4. $x(n) = 2x(n-1) + 4x(n-2), \quad x(0) = 1, \quad x(1) = 2$

Primary handwritten resolution:

Teil 1.4) $x(n) = 2x(n-1) + 4x(n-2)$ $x(0) = 1$ $x(1) = 2$ for $n \geq 1$

$x(n) = r^n$
 $r^n = 2r^{n-1} + 4r^{n-2}$
 $r^2 = 2r + 4$
 $r^2 - 2r - 4 = 0$
 $r = \frac{2 \pm \sqrt{4 + 16}}{2} = \frac{2 \pm \sqrt{20}}{2} = \frac{2 \pm 2\sqrt{5}}{2} = 1 \pm \sqrt{5}$
 $x(n) = C_1(1+\sqrt{5})^n + C_2(1-\sqrt{5})^n$

$x(0) = 1$
 $x(0) = C_1(1+\sqrt{5})^0 + C_2(1-\sqrt{5})^0 = 1$
 $C_1 + C_2 = 1$
 Substitute $C_2 = 1 - C_1$
 $C_1(1+\sqrt{5}) + (1-C_1)(1-\sqrt{5}) = 2$
 $C_1 + C_1\sqrt{5} + 1 - \sqrt{5} - C_1 + C_1\sqrt{5} = 2$
 $2C_1\sqrt{5} + 1 - \sqrt{5} = 2$
 $2C_1\sqrt{5} = 1 + \sqrt{5}$
 $C_1 = \frac{1+\sqrt{5}}{2\sqrt{5}}$
 $C_2 = 1 - \frac{1+\sqrt{5}}{2\sqrt{5}} = \frac{2\sqrt{5}}{2\sqrt{5}} - \frac{1+\sqrt{5}}{2\sqrt{5}} = \frac{\sqrt{5}-1}{2\sqrt{5}}$

Now find C_2 $C_2 = 1 - \frac{1+\sqrt{5}}{2\sqrt{5}} = \frac{2\sqrt{5}}{2\sqrt{5}} - \frac{1+\sqrt{5}}{2\sqrt{5}} = \frac{\sqrt{5}-1}{2\sqrt{5}}$

a) State and simplify the final closed form
 Substitute C_1 and C_2 back into the general solution
 $x(n) = \left(\frac{1+\sqrt{5}}{2\sqrt{5}}\right)(1+\sqrt{5})^n + \left(\frac{\sqrt{5}-1}{2\sqrt{5}}\right)(1-\sqrt{5})^n$

• We can pull out the common denominator $\frac{1}{2\sqrt{5}}$ and combine the terms. Note $(\sqrt{5}-1) = -(1-\sqrt{5})$
 we can pull the common denominator
 $x(n) = \frac{1}{2\sqrt{5}} [(1+\sqrt{5})^{n+1} - (1-\sqrt{5})^{n+1}]$
 $x(n) = \frac{1}{2\sqrt{5}} [(1+\sqrt{5})^{n+1} - (1-\sqrt{5})^{n+1}]$

Part 2) Verifying using Mathematical Induction
 $x(n) = \frac{1}{2\sqrt{5}} [(1+\sqrt{5})^{n+1} - (1-\sqrt{5})^{n+1}]$ holds true for all integer $n \geq 0$

Base case
 $n=0$
 $x(0) = \frac{1}{2\sqrt{5}} [(1+\sqrt{5})^1 - (1-\sqrt{5})^1] = \frac{1}{2\sqrt{5}} [2\sqrt{5}] = 1$ this matches $x(0) = 1$
 for $n \geq 1$
 $x(1) = \frac{1}{2\sqrt{5}} [(1+\sqrt{5})^2 - (1-\sqrt{5})^2]$
 $= \frac{1}{2\sqrt{5}} [(1+2\sqrt{5}+5) - (1-2\sqrt{5}+5)]$
 $= \frac{1}{2\sqrt{5}} [4\sqrt{5}] = 2$

Exercise 1.4 handwritten work: characteristic roots, constants, and base cases.

2) Inductive hypothesis - Assume the formula for $k \leq 0 \leq k \leq m$ where $m \geq 1$ asssom for.

$$x(m) = \frac{1}{2\sqrt{5}} [(1+\sqrt{5})^{m+1} - (1-\sqrt{5})^{m+1}]$$

$$x(m-1) = \frac{1}{2\sqrt{5}} [(1+\sqrt{5})^m - (1-\sqrt{5})^m]$$

3) Inductive step! We must show that the formula holds for $n=m+1$

$$x(m+1) = \frac{1}{2\sqrt{5}} [(1+\sqrt{5})^{m+2} - (1-\sqrt{5})^{m+2}]$$

Using the original recurrence relation

$$x(m+1) = 2x(m) + 4x(m-1)$$

$$\text{Substitute } x(m+1) = 2 \left(\frac{(1+\sqrt{5})^{m+1} - (1-\sqrt{5})^{m+1}}{2\sqrt{5}} \right) + 4 \left(\frac{(1+\sqrt{5})^m - (1-\sqrt{5})^m}{2\sqrt{5}} \right)$$

$$x(m+1) = \frac{1}{2\sqrt{5}} [2(1+\sqrt{5})^{m+1} + 4(1+\sqrt{5})^m - 2(1-\sqrt{5})^{m+1} - 4(1-\sqrt{5})^m]$$

$$\text{Notice that } (1+\sqrt{5})^2 = 1 + 2\sqrt{5} + 5 = 6 + 2\sqrt{5}$$

$$(1+\sqrt{5})^m [6 + 2\sqrt{5}] = (1+\sqrt{5})^m (1+\sqrt{5})^2 = (1+\sqrt{5})^{m+2}$$

Factor out $-(1-\sqrt{5})^m$ from the last two terms!!

$$= -(1-\sqrt{5})^m [2(1-\sqrt{5}) + 4] = -(1-\sqrt{5})^m [6 - 2\sqrt{5}]$$

$$\text{Notice that } (1-\sqrt{5})^2 = 6 - 2\sqrt{5}$$

$$\text{So, } -(1-\sqrt{5})^m [6 - 2\sqrt{5}] = -(1-\sqrt{5})^m (1-\sqrt{5})^2 = -(1-\sqrt{5})^{m+2}$$

Substitute these back into the main equation

$$x(m+1) = \frac{1}{2\sqrt{5}} [(1+\sqrt{5})^{m+2} - (1-\sqrt{5})^{m+2}]$$

Conclusion

the resulting expression for $m+1$ matches the structure of our closed-form solution by the principle of strong mathematical induction

Exercise 1.4 handwritten work: inductive step.

Typed solution for contrast and verification.

Part 1: Finding the Closed Form.

Assume $x(n) = r^n$. Substituting into the recurrence,

$$r^n = 2r^{n-1} + 4r^{n-2}.$$

Dividing by r^{n-2} gives

$$r^2 = 2r + 4 \implies r^2 - 2r - 4 = 0.$$

Using the quadratic formula,

$$r = \frac{2 \pm \sqrt{(-2)^2 - 4(1)(-4)}}{2} = \frac{2 \pm \sqrt{20}}{2} = 1 \pm \sqrt{5}.$$

Thus the general solution is

$$x(n) = c_1(1 + \sqrt{5})^n + c_2(1 - \sqrt{5})^n.$$

Now apply the initial conditions. For $n = 0$,

$$x(0) = c_1 + c_2 = 1 \implies c_2 = 1 - c_1.$$

For $n = 1$,

$$x(1) = c_1(1 + \sqrt{5}) + c_2(1 - \sqrt{5}) = 2.$$

Substituting $c_2 = 1 - c_1$,

$$\begin{aligned} 2 &= c_1(1 + \sqrt{5}) + (1 - c_1)(1 - \sqrt{5}) \\ &= c_1 + c_1\sqrt{5} + 1 - \sqrt{5} - c_1 + c_1\sqrt{5} \\ &= 1 - \sqrt{5} + 2c_1\sqrt{5}. \end{aligned}$$

Therefore,

$$2c_1\sqrt{5} = 1 + \sqrt{5} \implies c_1 = \frac{1 + \sqrt{5}}{2\sqrt{5}}.$$

Then

$$c_2 = 1 - c_1 = 1 - \frac{1 + \sqrt{5}}{2\sqrt{5}} = \frac{\sqrt{5} - 1}{2\sqrt{5}}.$$

So

$$x(n) = \frac{1 + \sqrt{5}}{2\sqrt{5}}(1 + \sqrt{5})^n + \frac{\sqrt{5} - 1}{2\sqrt{5}}(1 - \sqrt{5})^n.$$

Since $\sqrt{5} - 1 = -(1 - \sqrt{5})$, this can be written as the simplified closed form

$$x(n) = \frac{1}{2\sqrt{5}} \left[(1 + \sqrt{5})^{n+1} - (1 - \sqrt{5})^{n+1} \right]$$

Part 2: Verification by Strong Mathematical Induction.

We prove that

$$x(n) = \frac{1}{2\sqrt{5}} \left[(1 + \sqrt{5})^{n+1} - (1 - \sqrt{5})^{n+1} \right]$$

for all integers $n \geq 0$.

- **Base Cases:**

$$\begin{aligned}
 x(0) &= \frac{1}{2\sqrt{5}} \left[(1 + \sqrt{5}) - (1 - \sqrt{5}) \right] = \frac{2\sqrt{5}}{2\sqrt{5}} = 1, \\
 x(1) &= \frac{1}{2\sqrt{5}} \left[(1 + \sqrt{5})^2 - (1 - \sqrt{5})^2 \right] \\
 &= \frac{1}{2\sqrt{5}} \left[(6 + 2\sqrt{5}) - (6 - 2\sqrt{5}) \right] = \frac{4\sqrt{5}}{2\sqrt{5}} = 2.
 \end{aligned}$$

These match $x(0) = 1$ and $x(1) = 2$.

- **Inductive Hypothesis:** Assume the formula is true for all integers k such that $0 \leq k \leq m$, where $m \geq 1$. In particular,

$$\begin{aligned}
 x(m) &= \frac{1}{2\sqrt{5}} \left[(1 + \sqrt{5})^{m+1} - (1 - \sqrt{5})^{m+1} \right], \\
 x(m-1) &= \frac{1}{2\sqrt{5}} \left[(1 + \sqrt{5})^m - (1 - \sqrt{5})^m \right].
 \end{aligned}$$

- **Inductive Step:** We prove the formula for $m+1$. Using the original recurrence,

$$\begin{aligned}
 x(m+1) &= 2x(m) + 4x(m-1) \\
 &= \frac{2}{2\sqrt{5}} \left[(1 + \sqrt{5})^{m+1} - (1 - \sqrt{5})^{m+1} \right] \\
 &\quad + \frac{4}{2\sqrt{5}} \left[(1 + \sqrt{5})^m - (1 - \sqrt{5})^m \right] \\
 &= \frac{1}{2\sqrt{5}} \left[2(1 + \sqrt{5})^{m+1} + 4(1 + \sqrt{5})^m \right. \\
 &\quad \left. - 2(1 - \sqrt{5})^{m+1} - 4(1 - \sqrt{5})^m \right] \\
 &= \frac{1}{2\sqrt{5}} \left[(1 + \sqrt{5})^m (2(1 + \sqrt{5}) + 4) \right. \\
 &\quad \left. - (1 - \sqrt{5})^m (2(1 - \sqrt{5}) + 4) \right].
 \end{aligned}$$

Notice that

$$2(1 \pm \sqrt{5}) + 4 = 6 \pm 2\sqrt{5} = (1 \pm \sqrt{5})^2.$$

Substituting this into the previous expression,

$$x(m+1) = \frac{1}{2\sqrt{5}} \left[(1 + \sqrt{5})^{m+2} - (1 - \sqrt{5})^{m+2} \right],$$

which is exactly the required formula for $n = m+1$.

Therefore the formula is verified.

Exercise 2: Fibonacci Sequence Programs

The Fibonacci sequence is generated using the following recurrence: $T(n) = T(n-2) + T(n-1)$.

2.1. Generating the sequence using iteration

Source file: fibonacci_iterative.cpp

Only the most important fragment of the iterative program is shown below:

```
int prev2 = 0;
int prev1 = 1;
int current;

for (int i = 3; i <= n; i++) {
    current = prev1 + prev2;
    cout << current << " ";
    prev2 = prev1;
    prev1 = current;
}
```

In this fragment, `prev2` and `prev1` store the two previous Fibonacci values. At each iteration, the next term is obtained by adding them. Then the variables are updated so the program can continue generating the sequence without calling the function again.

Execution output:

```
PS C:\Users\User\Desktop\important\lab02> .\fibonacci_iterative.exe
Fibonacci Sequence (Iterative) up to 10 terms:
0 1 1 2 3 5 8 13 21 34
```

Captured output after compiling and running `fibonacci_iterative.cpp`.

2.2. Generating the sequence using recursion

Source file: fibonacci_recursive.cpp

Only the recursive rule and the sequence generation loop are shown:

```
if (n == 0) {
    return 0;
}
if (n == 1) {
    return 1;
}

return fibonacciRecursive(n - 1) + fibonacciRecursive(n - 2);
```


This fragment represents the recurrence relation directly. The cases `n == 0` and `n == 1` stop the recursion, while the last line computes each term using the two previous terms.

```
for (int i = 0; i < terms; i++) {  
    cout << fibonacciRecursive(i) << " ";  
}
```

This loop calls the recursive function for each position of the sequence and prints the corresponding Fibonacci value.

Execution output:

```
PS C:\Users\User\Desktop\important\lab02> .\fibonacci_recursive.exe  
Fibonacci Sequence (Recursive) up to 10 terms:  
0 1 1 2 3 5 8 13 21 34
```

Captured output after compiling and running `fibonacci_recursive.cpp`.